

CE 211: Civil Engineering Materials

Concrete

Concrete is the most widely used construction material.

Ingredients of Fresh Concrete

Portland cement

Water: Potable water

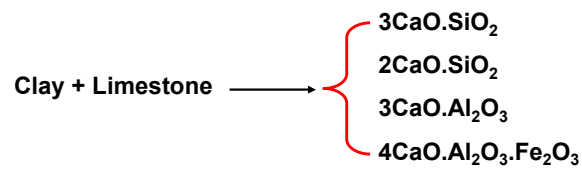
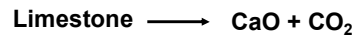
Aggregate: Coarse aggregate (gravel, crushed stone)
Fine aggregate (sand)

Mineral admixtures: Fly ash, Slag, Silica fume, Rice husk ash etc.

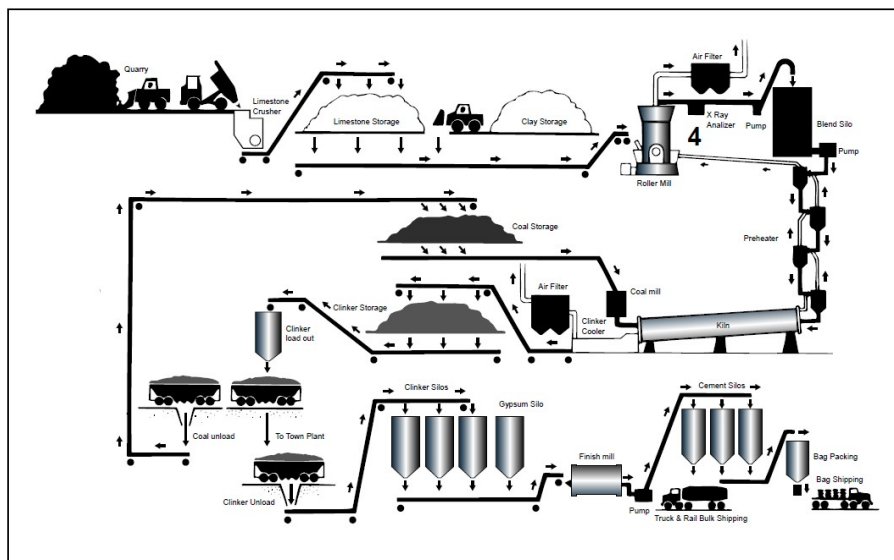
Chemical admixtures: Water reducing admixtures, Accelerating admixtures,
Retarding admixtures and Air entraining admixtures

Portland Cement

Chemical reactions, which are taking place in cement kiln may be expressed as follows:



Gypsum: approximately 5% is interground with clinker for controlling early setting and hardening behaviour of cement.



Flow diagram of dry process for Portland cement manufacture

Portland Cement

Approximate oxide composition limits of Portland cement

Oxide	Content (%)
CaO	60 – 67
SiO ₂	17 – 25
Al ₂ O ₃	3 – 8
Fe ₂ O ₃	0.5 – 6.0
MgO	0.1 – 4.0
Alkalis	0.2 – 1.3
SO ₃	1 – 3

Main compounds of Portland cement

1. Tricalcium Silicate (C₃S) $3\text{CaO} \cdot \text{SiO}_2$
2. Dicalcium Silicate (C₂S) $2\text{CaO} \cdot \text{SiO}_2$
3. Tricalcium Aluminate (C₃A) $3\text{CaO} \cdot \text{Al}_2\text{O}_3$
4. Tetracalcium Aluminoferrite (C₄AF) $4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$

C₃S and C₂S: most important compounds, which are responsible for the strength of hydrated cement paste.

Main compounds of Portland cement

Bogue's equations for theoretical or potential compound composition:

$$C_3S = 4.07(CaO) - 7.60(SiO_2) - 6.72(Al_2O_3) - 1.43(Fe_2O_3) - 2.85(SO_3)$$

$$C_2S = 2.87(SiO_2) - 0.754(3CaO.SiO_2)$$

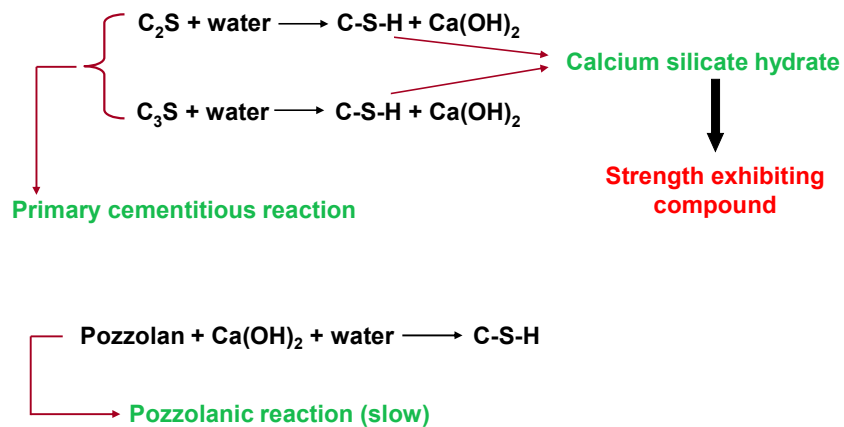
$$C_3A = 2.65(Al_2O_3) - 1.69(Fe_2O_3)$$

$$C_4AF = 3.04(Fe_2O_3)$$

The terms in the brackets represent the percentage of the given oxide in the total mass of cement.

Hydration of cement

Hydration of cement: an exothermic reaction.



Portland cement

Heat of hydration

- It is defined as the quantity of heat (in joules) per gram of unhydrated cement, liberated upon complete hydration at a given temperature.

Fineness of cement

- The fineness of cement affects its reactivity with water.
- Since, the hydration starts at the surface of the cement particles, the total surface area of cement represents the material available for hydration. Thus, the rate of hydration depends on the fineness of cement particles.
- For rapid development of strength, a higher fineness is required.
- For a given compound composition, the rate of reactivity and thus, the strength development can be enhanced by finer grinding of cement. However, the cost of grinding, and the heat liberated upon hydration set some limits on fineness of cement.

Types of Portland cement

Ordinary Portland cement (OPC)

Rapid hardening Portland cement (RHPC)

Portland pozzolana cement (PPC)

Portland slag cement (PSC)

Blended cements

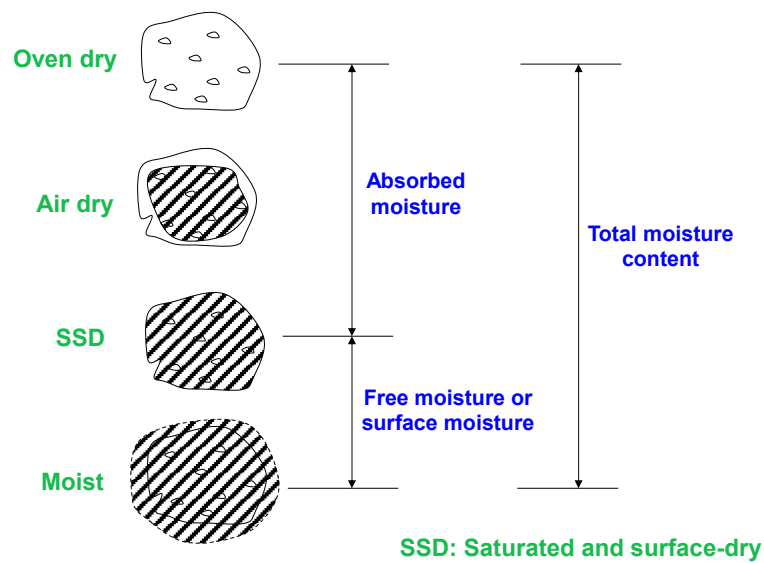
Sulphate resisting Portland cement (SRPC)

Aggregates

Coarse aggregate (gravel, crushed stone) and Fine aggregate (sand)

- Shape of aggregates
- Maximum size of aggregate (MSA)
- Nominal MSA
- Particle size distribution curve (Grading curve)
- Fineness modulus
- Specific gravity
- Bulk density
- Moisture content

Moisture in aggregate



Moisture in aggregate

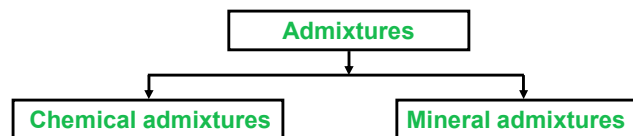
$$\text{Absorption (\%)} = \frac{(\text{SSD mass} - \text{oven dry mass})}{\text{Oven dry mass}} \times 100$$

SSD: Saturated and surface-dry

$$\text{Total moisture content (\%)} = \frac{(\text{Wet mass} - \text{oven dry mass})}{\text{Oven dry mass}} \times 100$$

$$\text{Surface moisture (\%)} = \text{Total moisture content (\%)} - \text{Absorption (\%)}$$

Admixtures



Chemical admixtures

- Water reducing admixtures
- Accelerating admixtures
- Retarding admixtures
- Air entraining admixtures

Mineral admixtures

- Natural pozzolans
- Industrial byproducts

Chemical admixtures

Water reducing admixtures

1. Increase the slump (indicator of workability) of concrete.
2. Increase the compressive strength of concrete for same slump.
3. The cost of concrete mix, which is primarily determined by the cement content, can be reduced.

Concrete Mix	Cement content (kg/m ³)	w/c ratio	Water content (kg/m ³)	Slump (mm)	Compressive strength (N/mm ²)	
					7 day	28 day
Base mix	300	0.62	186	50	25	37
Improved consistency	300	0.62	186	<u>100</u>	26	38
Increased compressive strength	300	0.56	168	50	<u>34</u>	<u>46</u>
Reduced cost	<u>270</u>	0.62	167.4	50	25.5	37.5

With water reducing admixture

Chemical admixtures

Water reducing admixtures

Conventional water reducing admixtures (Plasticizers):

- Decrease the water requirement by 5% - 15%.

High-range water reducing admixtures (Superplasticizers):

- Decrease the water requirement by 25% - 35%.

Types of Superplasticizers:

- Sulphonated melamine formaldehyde condensates
- Sulphonated naphthalene formaldehyde condensates
- Polycarboxylic ether-based

Chemical admixtures

Accelerating admixtures

- To accelerate setting and early strength development of concrete.
- Cold weather applications.
- To reduce curing time.
- Permit early removal of formwork.
- Calcium chloride is the most widely used accelerator.

Retarding admixtures

- Delay the setting and hardening of concrete.
- Useful when concreting in hot weather.
- In preventing formation of cold joints between successive lifts.
 - Sugar
 - Soluble zinc salts
 - Carbohydrate derivatives
 - Soluble borates

Chemical admixtures

Air entraining admixtures

- Dispersing a large number of microscopic air bubbles of diameter from 20 - 2000 μm .
- Result in disruption of the continuity of capillary pores.
- Reduce internal stresses caused by expansion of pore water upon freezing.
- Improvement in workability.
- Reduction in segregation tendency.
- Control of bleeding.

The admixtures may consist of:

- | | |
|-----------------------------|------------------------|
| - Salts of wood resins | - Petroleum acids |
| - Animal and vegetable fats | - Synthetic detergents |

Mineral admixtures

Natural pozzolans

- Volcanic tuff, Pumice, Diatomaceous earth etc.

Industrial byproducts

- Fly ash
- Ground granulated blast-furnace slag (GGBS)
- Silica fume
- Rice husk ash

- When used as part replacement for cement, the mineral admixtures are called as supplementary cementitious materials.

Fly ash

- Byproduct from burning of powdered coal in thermal power plants.
- Class F fly ash (CaO < 10%): from burning of bituminous coal.
- Class C fly ash (CaO: 10 - 30%): burning of subbituminous coal or lignite.

Mineral admixtures

Fly ash

- Class F fly ash (low-calcium fly ash): possesses pozzolanic property and little or no cementing property.
- Class C fly ash (high-calcium fly ash): possesses pozzolanic property and some cementing property.

Benefits of adding fly ash to concrete:

- Increase in ultimate strength (later-age strength)
- Improved workability and pumpability
- Reduced temperature rise
- Reduced permeability
- Reduced alkali-aggregate reaction
- Improved resistance to sulfate attack
- Economy

Mineral admixtures

Ground granulated blast-furnace slag (GGBS)

- Granulated blast-furnace slag is a waste product in the manufacture of pig iron.
- Blast-furnace slag is a non-metallic product consisting essentially of silicates and aluminates of calcium and other bases.
- Lowers heat of hydration.
- Decreases permeability of concrete.

Silica fume

- Byproduct from the electric arc furnaces of the silicon metal and ferrosilicon industries.
- Silica fume consists of 85 - 95% of noncrystalline silicon dioxide.
- Particles of about 0.1 μm in size.
- Silica fume is a very highly pozzolanic material.
- For producing high strength and durable concrete.

Mineral admixtures

Rice husk ash

- Rice husk: produced during dehusking operation of paddy rice.
- Rice husks on combustion produce rice husk ash.
- Rice husk ash consists essentially of pure silica in noncrystalline form.
- The strength and durability aspects can be improved by adding rice husk ash to concrete.

Workability of concrete

Workability- a composite property with following components:

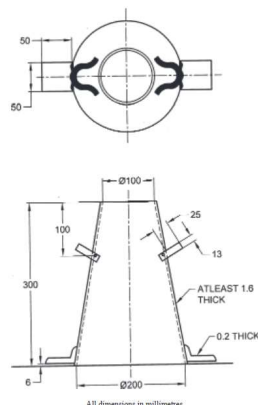
- Consistency: describes the ease of flow.
- Cohesiveness: describes the stability (or lack of bleeding and segregation characteristics).

Tests for workability:

- Slump test
- Vebe test
- Compacting factor test

Slump test

IS : 1199 (Part 2) – 2018



NOTE — To facilitate the lifting of the mould in a vertical direction, it is recommended that suitable guide attachments may be provided. Any rods used in the construction of the mould shall be countersunk flush on the inside of the cone. Attachments shall be preferably welded to the mould.

FIG. 1 TYPICAL Mould FOR SLUMP TEST

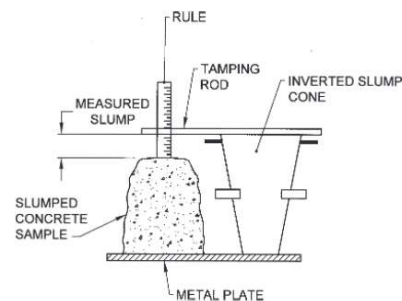


FIG. 2 SLUMP MEASUREMENT

Slump test

IS : 1199 (Part 2) – 2018

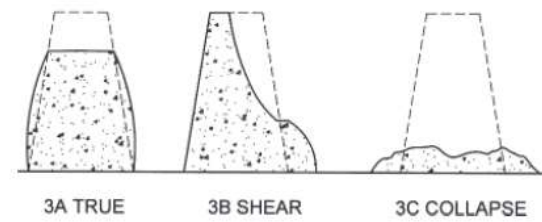
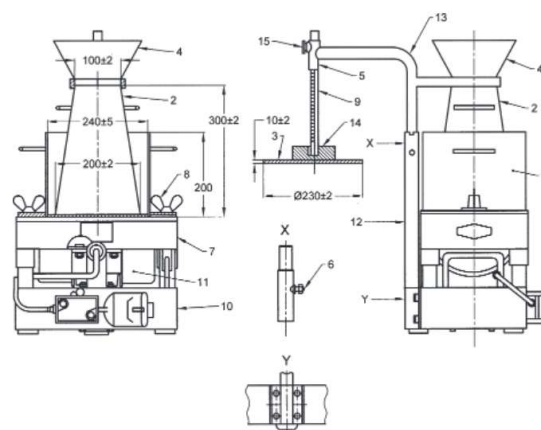


FIG. 3 TYPES OF SLUMP

Vebe test

IS : 1199 (Part 2) – 2018



Key		
1 container	6 set-screw	11 vibrator unit
2 mould	7 vibrating table	12 holder
3 transparent disc	8 wing nuts	13 swivel arm
4 funnel	9 rod	14 weight
5 guide sleeve	10 base for vibrator	15 screw

All dimensions in millimetres.
FIG. 4 VEE BEE CONSISTOMETER

Compacting factor test

IS : 1199 (Part 2) – 2018

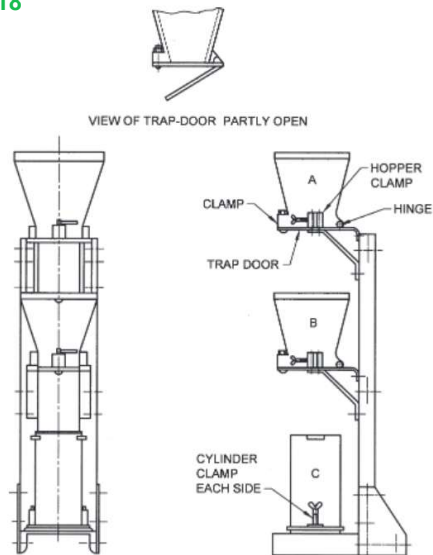


FIG. 5 COMPACTION FACTOR APPARATUS

Workability of concrete

Factors affecting workability:

- Water content
- Aggregate type and grading
- Aggregate/cement ratio
- Presence of admixtures
- Fineness of cement
- Time
- Temperature

Fresh concrete

Segregation

- It is defined as the separation of the constituents of a heterogeneous mix so that their distribution is no longer uniform.
- In concrete, the differences in the size of the particles and sometimes, the differences in the specific gravity of the mix constituents are the primary cause of segregation. However, the extent of segregation can be controlled by the choice of suitable grading and by care in handling.

Two forms of segregation:

- In the first form of segregation, the coarser particles tend to separate, and it occurs in lean concrete mixes, if the mix is too dry.
- The second form of segregation occurs particularly in wet concrete mixes where the grout separates from the mix.

Bleeding

- It is also known as water gain. This is a form of segregation where some of the water in the mix tends to rise to the surface of the freshly placed concrete.
- Bleeding is caused by the inability of the solid constituents of the mix to hold all the mixing water when they settle downwards.

References

- Mehta, P. K., and Monteiro, P. J. M. Concrete: Microstructure, Properties, and Materials, Tata McGraw-Hill, Third edition, 2006.
- Neville, A. M., and Brooks, J. J. Concrete Technology, Pearson Education, Fourth Indian reprint, 2004.
- Somayaji, S. Civil Engineering Materials, Prentice Hall, Second edition, 2001.
- Mamlouk, M. S., and Zaniewski, J. P. Materials for Civil and Construction Engineers, Pearson, Prentice Hall, Second edition, 2006.
- IS 1199 (Part 2) : 2018, Fresh concrete – Methods of sampling, testing and analysis, Part 2: Determination of consistency of fresh concrete, Bureau of Indian Standards, New Delhi.